

A deletion bridge for the off-hook order law:
split-independence and a rank-one collapse reduce $\text{ord} = D$
to a single-vector reach bound
(the family $\lambda = (2, 2, 1^m)$, uniformly in m)

Clio

31 May 2026 (prove session)

Abstract

The off-hook order law asserts $\text{ord}_{x=q^2} Z_\lambda = D := \binom{m}{2}$ for $\lambda = (2, 2, 1^m)$, equivalently the *split-independent per-subset vanishing* $M(S) \equiv 0$ on V^λ for every insertion subset S with $|S| < D$. Prior work proved the lower bound $\text{ord} \geq \tau$ ($\tau = m - 1$) but stalled on the split: the per-block deficit count $g(t) = t + \binom{\tau-t+1}{2}$ is loose ($< D$) whenever the number $t = |S_T|$ of tail insertions is positive, and computation showed the missing killing comes from the *placement* of insertions, not their count. We remove the split entirely. Writing each inserted projector as $P_{-1} = I - P_q$ and expanding the ordered product, Möbius inversion on the Boolean lattice of word positions yields the **deletion bridge**

$$M(S) = \sum_{U \subseteq S} (-1)^{|S|-|U|} \Pi_{\text{del}}(U), \quad \Pi_{\text{del}}(U) = \sum_{V \subseteq U} M(V),$$

where $\Pi_{\text{del}}(U)$ is the all- P_q staircase Ω with the factors in U *deleted* (set to I). Two consequences are immediate and uniform in m : (i) $\text{ord} = d_{\text{del}} := \min\{|U| : \Pi_{\text{del}}(U) \neq 0\}$; and (ii) *the split-independent per-subset order law is equivalent to the single split-free statement* $\Pi_{\text{del}}(U) = 0$ for all $|U| < D$ — because each $M(S)$ with $|S| < D$ is a $\pm$ -combination of $\Pi_{\text{del}}(U)$ with $|U| \leq |S| < D$. This dissolves the intermediate-split and heavy-tail regimes that obstructed the insertion approach. We then show computationally (exhaustively for $m = 3$, by sampling for $m = 4$) that for $|U| \leq D$ the deletion operator *collapses through the two longest blocks to rank* ≤ 1 : the tail sends V^λ to a line $\langle w \rangle$, and the prefix is a single-vector walk on w . The deletion order law thereby reduces to a **single-vector reach bound**, the exact off-hook twin of the unconditional hook chip-firing bound $\binom{m}{2}$ proved for $(2, 1^m)$. We isolate the remaining crux — a cost-to-go inequality $\Psi(w) \geq D - |S_T|$ on the deposited line — and show by example that it has no support-only potential (the obstruction is the same no-cancellation phenomenon as the hook $G_{\binom{m}{2}} \neq 0$ gap). All identities are verified by exact rational arithmetic ($q = 5/7$, cross-checked $q = 2/3$) for $m \leq 4$.

1 Setup

We retain the conventions of the 29–30 May deficit papers. $H_n(q)$ is the Iwahori–Hecke algebra, $T_i^2 = (q - 1)T_i + q$; on the Hoefsmit seminormal module V^λ set $S_i := T_i + 1$ and

$$E_i^+ := \ker(T_i - q)|_{V^\lambda} = \text{im } P_q^{(i)}, \quad E_i^- := \ker(T_i + 1)|_{V^\lambda} = \text{im } P_{-1}^{(i)},$$

with $P_q^{(i)} = (T_i + 1)/(q + 1)$ and $P_{-1}^{(i)} = (qI - T_i)/(q + 1)$ the orthogonal eigenprojectors, $P_q^{(i)} + P_{-1}^{(i)} = I$.

The staircase word and the per-subset operators. For $\lambda = (2, 2, 1^m)$ we have $n = m + 4$, $N = \binom{n}{2}$, $\tau := \tau(\lambda) = m - 1$, and

$$D := \frac{\tau(\tau+1)}{2} = \binom{m}{2}.$$

Write $B_b := S_b S_{b-1} \cdots S_1$ for the descending block of length b (generators $1, \dots, b$), $b = 1, \dots, n-1$. The staircase word is the concatenation $B_1 B_2 \cdots B_{n-1}$; let $(i_1, i_2, \dots, i_N)$ be its letter sequence. We multiply projector choices left to right as matrices, so the rightmost letter acts *first* on a vector and the longest block B_{n-1} acts first. For a subset $S \subseteq \{1, \dots, N\}$ of word positions define the *insertion* operator

$$M(S) := \prod_{k=1}^N A_k, \quad A_k = P_{-1}^{(i_k)} \ (k \in S), \quad A_k = P_q^{(i_k)} \ (k \notin S),$$

and the *deletion* operator

$$\Pi_{\text{del}}(U) := \prod_{k=1}^N C_k, \quad C_k = I \ (k \in U), \quad C_k = P_q^{(i_k)} \ (k \notin U).$$

Thus $\Pi_{\text{del}}(\emptyset) = \prod_k P_q^{(i_k)} =: \Omega$ is the all- P_q degenerate staircase, and $\Pi_{\text{del}}(U)$ is Ω with the factors indexed by U removed. As established (operator reformulation of the order law),

$$\text{ord}_{x=q^2} Z_\lambda = \min\{|S| : M(S) \neq 0 \text{ on } V^\lambda\} =: \text{ord}.$$

The order law is the assertion $\text{ord} = D$, and its operator form is the *split-independent per-subset vanishing*: $M(S) \equiv 0$ for all $|S| < D$, with no condition on how S distributes along the word.

Depth. For $W \subseteq V^\lambda$ put $\text{dep}(W) := \max\{d \geq 0 : W \subseteq \bigcap_{j=1}^d E_j^-\}$.

The obstruction we remove. The prior insertion analysis splits $S = S_R \sqcup S_T$ into prefix $(B_1 \cdots B_{n-2})$ and tail $(B_{n-2} B_{n-1})$ positions and proves, for $|S_T| < \tau$, the deficit bound $M(S) \neq 0 \Rightarrow |S| \geq g(|S_T|)$ with $g(t) = t + \binom{\tau-t+1}{2}$. Since $g(0) = D$ but $g(t) < D$ for $t \geq 1$, this gives $\text{ord} \geq \tau$ but not $\text{ord} = D$; the residual is exactly the placement-sensitive “intermediate split” $1 \leq |S_T| < \tau$ together with the heavy-tail regime $|S_T| \geq \tau$.

2 The deletion bridge

Theorem 1 (Deletion bridge). *As operators on V^λ (indeed on any module), for every $S \subseteq \{1, \dots, N\}$,*

$$M(S) = \sum_{U \subseteq S} (-1)^{|S|-|U|} \Pi_{\text{del}}(U), \quad \text{and inversely} \quad \Pi_{\text{del}}(U) = \sum_{V \subseteq U} M(V). \quad (1)$$

Proof. Fix S . In the ordered product defining $M(S)$, the factor at a position $k \in S$ is $P_{-1}^{(i_k)} = I - P_q^{(i_k)}$, and at $k \notin S$ it is $P_q^{(i_k)}$. Expanding each factor $I - P_q^{(i_k)}$ ($k \in S$) as a sum of two terms, and keeping every position in its original slot (no reordering), we obtain

$$M(S) = \prod_{k \in S} (I - P_q^{(i_k)}) \prod_{k \notin S} P_q^{(i_k)} = \sum_{U \subseteq S} \left(\prod_{k \in U} I \right) \left(\prod_{k \in S \setminus U} (-P_q^{(i_k)}) \right) \prod_{k \notin S} P_q^{(i_k)},$$

where for each $U \subseteq S$ we chose the I -term at the positions of U and the $-P_q^{(i_k)}$ -term at the positions of $S \setminus U$. The sign is $(-1)^{|S \setminus U|} = (-1)^{|S| - |U|}$, and the surviving operator has P_q at every position except U (where it has I), i.e. it is $\Pi_{\text{del}}(U)$. This is the first identity. The second is Möbius inversion on the Boolean lattice: $\sum_{V \subseteq U} M(V) = \sum_{V \subseteq U} \sum_{W \subseteq V} (-1)^{|V| - |W|} \Pi_{\text{del}}(W) = \sum_{W \subseteq U} \Pi_{\text{del}}(W) \sum_{V: W \subseteq V \subseteq U} (-1)^{|V| - |W|} = \Pi_{\text{del}}(U)$, the inner alternating sum vanishing unless $W = U$. $\square$

Corollary 2 (Order law = deletion order law). *With $d_{\text{del}} := \min\{|U| : \Pi_{\text{del}}(U) \neq 0\}$, one has $\text{ord} = d_{\text{del}}$.*

Proof. If $M(S) \neq 0$ then by (1) some $\Pi_{\text{del}}(U) \neq 0$ with $U \subseteq S$, so $d_{\text{del}} \leq |U| \leq |S|$; minimising over S gives $d_{\text{del}} \leq \text{ord}$. If $\Pi_{\text{del}}(U) \neq 0$ then some $M(V) \neq 0$ with $V \subseteq U$, so $\text{ord} \leq |V| \leq |U|$; minimising gives $\text{ord} \leq d_{\text{del}}$. $\square$

Corollary 3 (Split-independence for free). *The following are equivalent, uniformly in m :*

- (a) $M(S) \equiv 0$ on V^λ for every insertion subset S with $|S| < D$ (the split-independent per-subset order law);
- (b) $\Pi_{\text{del}}(U) \equiv 0$ on V^λ for every deletion subset U with $|U| < D$ (the split-free deletion order law).

Proof. (b) $\Rightarrow$ (a): if $|S| < D$ then every $U \subseteq S$ has $|U| \leq |S| < D$, so each term in $M(S) = \sum_{U \subseteq S} (-1)^{|S| - |U|} \Pi_{\text{del}}(U)$ vanishes, whence $M(S) = 0$. (a) $\Rightarrow$ (b): symmetric, via $\Pi_{\text{del}}(U) = \sum_{V \subseteq U} M(V)$. $\square$

Remark 4 (Why this is the right move). Statement (a) carries the entire split apparatus — the cases $S_{\text{T}} = \emptyset$, $1 \leq |S_{\text{T}}| < \tau$, $|S_{\text{T}}| \geq \tau$ all had to be handled separately, and the middle case was where the count bound $g(t)$ went slack. Statement (b) has *no split*: U is an arbitrary subset of the whole word. Corollary 3 says the split was never essential — it was an artefact of writing the order law with the projectors P_{-1} rather than with $I - P_q$. The honest object is the single quantity d_{del} , the minimum number of factors one must delete from the degenerate staircase Ω to revive it on V^λ .

Corollary 5 (Recovering $\text{ord} \geq \tau$, split-free). $d_{\text{del}} \geq \tau$, hence $\text{ord} \geq \tau$; equivalently $\Pi_{\text{del}}(U) \equiv 0$ for $|U| < \tau$.

Proof. By Corollary 2 and the proved bound $\text{ord} \geq \tau$. Alternatively, directly: the deletion analogue of the deficit lemmas applies verbatim with I in place of P_{-1} . A block O_b acting on a depth- d subspace W kills it unless its first d letters are deleted (each $P_q^{(j)}$, $j \leq d$, annihilates $W \subseteq E_j^-$, while I fixes it), and the remaining letters — generators $\geq d + 1$ — commute past generators $\leq d - 1$ (axial gap ≥ 2) and preserve depth $d - 1$. Iterating over the prefix blocks below the tail's deposited depth gives the same arithmetic series, with $\geq \tau$ deletions forced before extinction can be avoided. $\square$

3 Rank-one collapse: the deletion law is a single-vector walk

By Corollary 3 it remains to study $\Pi_{\text{del}}(U)$ for $|U| < D$, an arbitrary subset of the whole word. The decisive structural fact is that for few deletions the operator factors through a line.

Proposition 6 (Rank-one collapse, computational). *For $\lambda = (2, 2, 1^m)$ and every U with $|U| \leq D$, the running image after the two longest blocks $B_{n-1}B_{n-2}$ (the tail) has rank ≤ 1 :*

$$\text{rank}(O_{n-2}^U O_{n-1}^U V^\lambda) \leq 1,$$

where O_b^U is block B_b with its U -positions deleted. Hence for $|U| \leq D$ the prefix $B_1 \cdots B_{n-3}$ acts on a line, and $\Pi_{\text{del}}(U) \neq 0$ iff the prefix walk keeps that line alive.

Status. Verified by exact arithmetic: exhaustively over all U with $|U| \leq D$ for $m = 3$ (where the all- P_q rank profile of $(B_{n-1}, \dots, B_1)$ is $(4, 1, 0, 0, 0, 0)$), and by random sampling for $m = 4$ (profile $(5, 1, 0, \dots)$). We expect Proposition 6 m -uniformly from the rank-one pinch (for $|a - b| \geq 2$, $P_q^{(a)} P_q^{(b)}$ is the projector onto a 1-dimensional joint q -eigenspace, proved for the whole family) together with the forward backbone collapse (the all- P_q prefix has image a single seminormal line, proved m -uniformly). A clean m -uniform proof that few deletions cannot defeat enough of these pinches is the natural next lemma; we treat the collapse as a hypothesis for the reduction below.

Corollary 7 (Reduction to a single-vector reach bound). *Assume Proposition 6. Split $U = U_T \sqcup U_R$ into tail and prefix positions, and for a deposited line $\langle w \rangle = O_{n-2}^{U_T} O_{n-1}^{U_T} V^\lambda$ let*

$$\Psi(w) := \min\{|U_R| : \text{the prefix with deletions } U_R \text{ keeps } \langle w \rangle \text{ nonzero}\}$$

be the cost-to-go. Then the deletion order law $d_{\text{del}} \geq D$ is equivalent to the single-vector reach bound

$$\Psi(w) \geq D - |U_T| \quad \text{for every tail-deposited line } w. \quad (2)$$

Proof. If $\Pi_{\text{del}}(U) \neq 0$ with $U = U_T \sqcup U_R$, the prefix keeps the deposited line alive, so $|U_R| \geq \Psi(w) \geq D - |U_T|$ by (2), giving $|U| \geq D$. Conversely, taking $U_T = \emptyset$, (2) reads $\Psi(w_0) \geq D$ for the all- P_q deposited line w_0 , which is exactly $d_{\text{del}} \geq D$ on the all- P_q -tail slice; the displayed inequality for $|U_T| \geq 1$ supplies the remaining slices. $\square$

The inequality (2) is the off-hook twin of the unconditional chip-firing lower bound for the hook $(2, 1^m)$, where $P_q^{(i)}$ is rank one on $V^{(2, 1^m)}$, the trace factorises into scalar sandwiches, and a reach principle (a descending run raises the reachable maximum by at most one) gives $\text{ord} \geq \binom{m}{2}$ with no positivity input. After the collapse of Proposition 6 the off-hook prefix is likewise a single-vector walk, and we expect the same reach principle.

4 The crux, and why no support-only potential closes it

The tautological route to (2) fails: the cost-to-go Ψ satisfies the super-triangle inequality $\Psi(O_b^d \langle v \rangle) \geq \Psi(\langle v \rangle) - |d|$ for any block deletion pattern d (immediate from its definition as a minimum), but this gives no *lower* bound on Ψ of the deposited line without an independent handle. One therefore seeks a *closed-form potential* φ on lines with

$$(i) \ \varphi(w) \geq D - |U_T| \quad \text{on tail-deposited lines,} \quad (ii) \ \varphi(O_b^d \langle v \rangle) \geq \varphi(\langle v \rangle) - |d|,$$

so that summing (ii) over the prefix blocks yields $|U_R| \geq \varphi(w) \geq D - |U_T|$.

The natural candidate, and its failure. Each seminormal basis vector of V^λ corresponds to its column–1 pair (a, b) , $a < b$, the entries of the two cells $(0, 1), (1, 1)$; the row–superstandard target line is $(a, b) = (2, 4)$. The graph distance to the target, $\text{ch}(a, b) := (a - 2) + (b - 4)$, gives a valid *lower* bound $\Psi(w) \geq \min_{(a,b) \in \text{supp}(w)} \text{ch}(a, b)$ on every line tested ($m = 3$). But this minimum is *not* $\geq D - |U_T|$: at $m = 3$, $|U_T| = 2$, the deposited line with support $\{(2, 4), (2, 5), (3, 4), (3, 5)\}$ has $\min \text{ch} = 0$ (the target itself lies in the support), yet $\Psi = 1 = D - |U_T|$. The cheap $(2, 4)$ –component is present in the support but *cancels* along the cheapest prefix walk, so the support alone does not determine Ψ .

Identification of the obstruction. Two distinct residuals meet here. The *lower* bound (2) is obstructed by the off–hook “merged–arc” (merged junction) gap: when few deletions break a shared pinch, a support–based reach principle *undershoots* the true cost — exactly as $\min_{(a,b) \in \text{supp}} \text{ch}$ gives 0 on a line whose true Ψ is 1. So a coefficient–blind reach principle does not suffice for the lower bound; one needs either a refined reach that charges merged pinches correctly, or a coefficient–aware argument. (Separately, the *equality* $\text{ord} = \binom{m}{2}$ rather than $\geq$ is the no–cancellation residual $G_{\binom{m}{2}} = \sum_{|S|=\binom{m}{2}} M(S) \neq 0$, already in hand from the survivor census.) The single–vector reach bound (2) is therefore not a function of the line’s $E_\bullet^-$ signature or of its support; we record it as the irreducible core of the lower bound.

Conjecture 8 (Single–vector reach bound). *For $\lambda = (2, 2, 1^m)$ the inequality (2) holds: every tail–deposited line w needs at least $D - |U_T|$ prefix deletions to survive. Equivalently $d_{\text{del}} \geq D$, equivalently (Corollary 3) the split–independent order law $\text{ord} = D$.*

5 Verification

All checks use the Hoefsmit seminormal machinery; structural identities are exact rational ($q = 5/7$, cross–checked $q = 2/3$), bulk screens use a generic rational q to exclude accidental cancellation. Scripts in `~/projects/scratch/2026-05-31-intermediate-split/`.

- *Bridge (1) as matrices.* Both identities hold exactly for randomly chosen S, U at $m = 3$ and $m = 4$.
- *Deletion order law $d_{\text{del}} = D$.* $m = 2$: $d_{\text{del}} = 1 = D$. $m = 3$: all $|U| \leq 2$ deletions give $\Pi_{\text{del}}(U) = 0$ exactly (1 + 21 + 210 subsets), first survivor at $|U| = 3 = D$. $m = 4$: all $|U| \leq 5$ give 0 (generic– q screen with early zero short–circuit), survivor at $|U| = 6 = D$.
- *Rank–one collapse.* For all U with $|U| \leq D$ at $m = 3$ the running rank after the tail is ≤ 1 ; likewise on 60 random subsets at $m = 4$. The all– P_q rank profiles are $(4, 1, 0, \dots)$ ($m = 3$) and $(5, 1, 0, \dots)$ ($m = 4$).
- *Cost–to–go.* For every tail–deposited line at $m = 3$, $\Psi(w) \geq D - |U_T|$ holds (minimum over deposits at fixed $|U_T|$ is attained with equality, giving the $D - |U_T|$ survivors); and Ψ is not a function of the $E_\bullet^-$ signature (signature $(1, 2)$ occurs with $\Psi \in \{3, 5\}$).

λ	m	τ	D	d_{del} (computed)	first survivor $ U $
$(2, 2, 1, 1)$	2	1	1	1	1
$(2, 2, 1, 1, 1)$	3	2	3	3	3
$(2, 2, 1, 1, 1, 1)$	4	3	6	6	6

6 Status

Proved (uniformly in m).

- Theorem 1 (the deletion bridge) — an elementary but exact operator identity, valid on any module.
- Corollary 2: $\text{ord} = d_{\text{del}}$.
- Corollary 3: the split-independent per-subset order law is *equivalent* to the split-free deletion order law $\Pi_{\text{del}}(U) \equiv 0$ for $|U| < D$. This is the structural payoff: it eliminates the intermediate-split and heavy-tail casework.
- Corollary 5: $d_{\text{del}} \geq \tau$ (hence $\text{ord} \geq \tau$), now split-free.

Reduced (modulo Proposition 6). The deletion order law $d_{\text{del}} \geq D$ is equivalent to the single-vector reach bound (2) (Corollary 7). Proposition 6 (the rank-one collapse through the tail) is verified for $m \leq 4$ and expected m -uniformly from the rank-one pinch and the forward backbone collapse.

Open (the crux). Conjecture 8: the single-vector reach bound $\Psi(w) \geq D - |U_T|$. We have shown it is not a function of the line's support or eigenspace signature; the obstruction to this *lower* bound is the off-hook merged-arc gap (a support-based reach undershoots when a shared pinch is broken). The reduction to a *single* split-free inequality on a *single* vector — the off-hook twin of the proved unconditional hook chip-firing bound $\binom{m}{2}$ — is the contribution of this note; closing it is the remaining work.